

Barkai Lazimy

Engineering Student | University of Wisconsin-Madison

PORTFOLIO WEBSITE | <https://bitarmada.github.io/Portfolio/>

EXPERIENCE

UW Madison Department of Food Science, Madison, WI — Research Intern

June 2024 - September 2024

Gained experience in a professional research environment, growing bacterial cultures, sequencing DNA, and running PCR reactions. Discovered and named several previously unknown strains of bacteria.

The Great Dane, Madison, WI — Expo / Servers Assistant

May 2025 - September 2025

Worked with customers and kept track of orders in a fast-paced environment.

Wisconsin Union Catering, Madison, WI — Student Caterer

June 2023 - September 2023

Prepared food, set up venues, and served guests at various events.

AIAlgo, Madison, WI — Website and backend developer

September 2022 - December 2022

Coded a website with custom UI and 3d animations using HTML, JS, and CSS. Created a secure account system and customer database from scratch to handle payments and subscriptions.

INDEPENDENT PROJECTS — MORE IN PORTFOLIO

Topology Optimization for Fluid Flows — AI-based algorithm

Created an algorithm for optimizing solid topologies in fluid flows, a problem that traditional optimization algorithms struggle to solve. I used a Convolutional Neural Network (CNN) to approximate fluid forces and Python + Pytorch/Tensorflow to compute topological gradients using back propagation.

AI Security System — Prototype and Business Plan

A prototype for a security product that utilizes non-lethal lasers and audio to deter potential threats. I designed and 3d printed a camera turret capable of locking onto attackers with guns/knives using an Arduino and a custom computer vision algorithm. The project won first place in the state for the DECA Entrepreneurship Innovation Plan 2025.

Electrochemical Carbon Capture Device — Prototype

Designed and built a low-cost carbon-capture device utilizing a novel electrochemical process to recycle sodium ions using a porous ceramic membrane. I proved the concept's viability by building a prototype from commonly available materials and testing CO2 concentrations.

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SKILLS

Programming

languages: JavaScript, Java, C, C++, Python, R, HTML/CSS, assembly, WASM, GPU shaders

AI and machine learning algorithms:

Optimization from scratch, Neuroevolution of Augmenting Topologies (NEAT), GANs, CNNs, and custom computer vision models. Experience with Pytorch and Tensorflow.

GPU Programming:

webGPU custom compute shaders WGSL and JS, WebGL and OpenGL graphics programming with GLSL.

Computer-Aided Design:

3d modeling and 3d printing experience. Topology optimization for fluid flow.

Teamwork:

I have experience working collaboratively and have strong written and verbal communication skills.



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